Scenarios

For each of the following scenarios describe the primary aerodynamics responsible acting on the aircraft. What went wrong? Which parts of the plane design were responsible? What should the team change for the next iteration? There's a lot of text in these paragraphs – not everything written is primarily responsible! There should be one or two crucial parameters.

1. There are heavy winds in Kansas today, and everyone is waiting on the flight line. The plane glistens in the sun. It has a long chord for its smaller wingspan, and surprisingly, features a T-tail empennage. The nosecone is short, and on its front is a large, heavy motor. You see the judges lining up, checking to see if the plane can take off in 20ft. Your pilot's nervous as a direct tailwind strikes the aircraft but begins to rapidly advance throttle. The long tail almost strikes the pavement; barely making it, the aircraft lifts off the ground. Just as the landing gear picks up, the plane begins to roll left and yaw into its turn. Just a few seconds into the flight, and the tendency gets worse. The plane hits the ground nose-first and crashes.

2. It's a cold February morning as the team gathers on the flying field. Everything's been checked: batteries charged, control surfaces deflect, "all packs are secure in the aircraft". Mitch turns up the throttle, and the motors spool up slowly. The tail dragger plane starts barreling down the runway as the tail wheel comes up. Mitch pulls up, but the plane stays on the ground. He tries moving the rudder, maneuvering the plane across the runway but there's no strong effect. The plane remains level. Finally, the landing gear hits a rock in the grass; the wide wing catches on the ground, and the plane rolls, yaws, then tips over.

3. Late august strikes and DBF is hosting a welcome back social at the flight club. Ishan is flying the Piper Cub after a summer of practice. The plane, which has a high aspect ratio, is up in the air with a couple hockey pucks as cargo. The team performed a CG check before flight; it was at 30% chord length. When he was taking off, he noticed he was pulling the stick up a lot more than usual. Also in the air, he was having trouble climbing. Cruising fast on a straightaway, Ishan performs quick wing-wave roll. Simulating a competition loop, he then banks rightwards and pulls on the elevator. The turn is wide and the plane struggles to keep it. Having to cough, Ishan lets go of the stick and the plane immediately pitches down. He quickly takes back over the controls and recovers it. Upon landing with the flaps lowered, he's holding the elevator all the way up. Still, he's sinking fast and hits the ground hard.

Challenges

These exercises are meant to be much more open-ended and creative. There are many possible approaches and even more infeasible ones. Start with a baseline design you know works, and bit by bit change it to push forward. Good luck!

1. Design an aircraft with the smallest possible empennage while maintaining stability in all modes. Your wing is constrained to our competition plane's dimensions.

2. What is the fastest possible equilibrium velocity you can achieve with any aircraft design? The aircraft must still be considered reasonably flyable.

3. Design the longest aircraft possible (nose to tail) while maintaining stability in all modes.